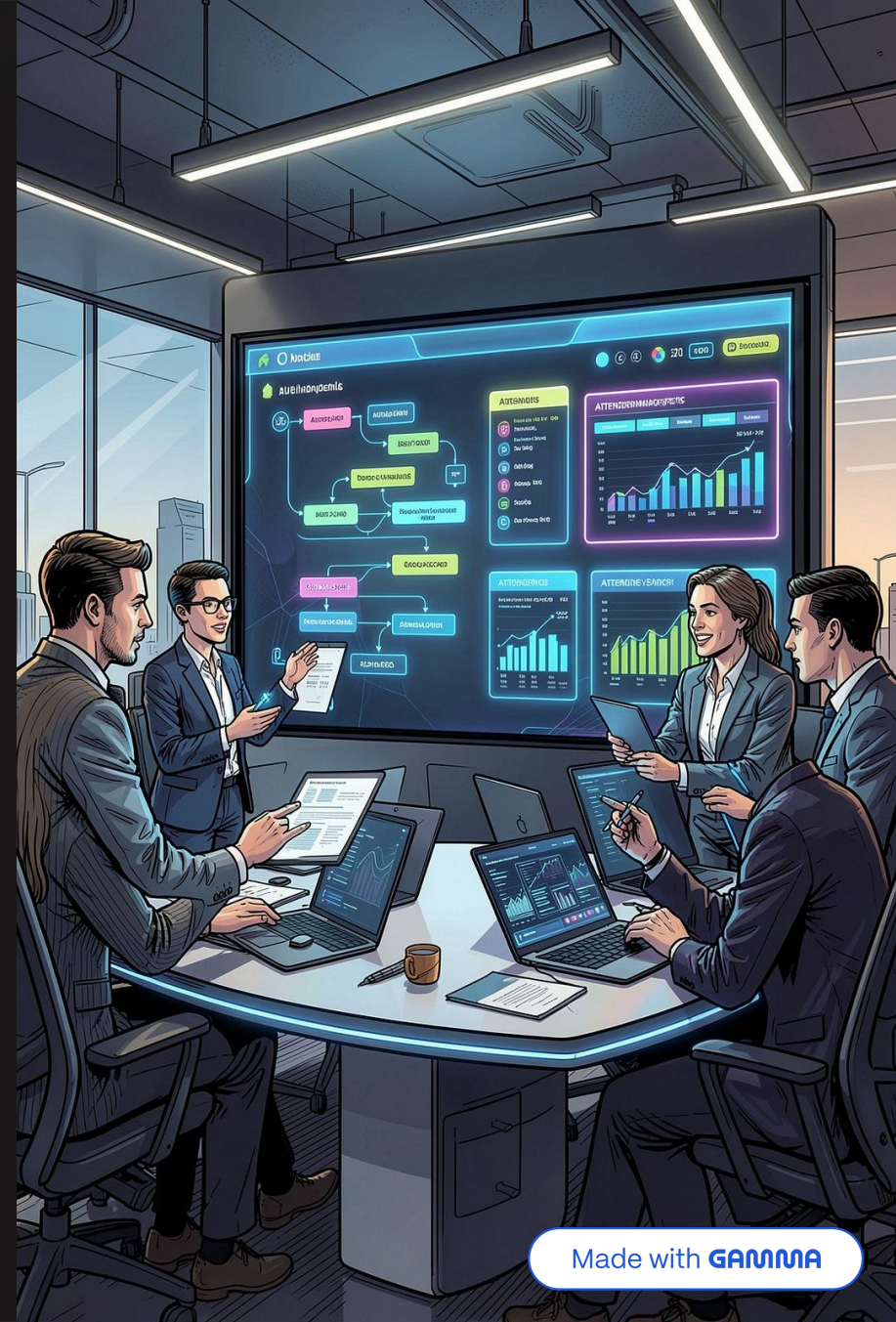


AI-Powered Event Management Platform

An End-to-End Automation System Built on n8n

N8N CAPSTONE PROJECT · SUMMER SCHOOL '26

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Manual Event Operations Don't Scale

Organizations running conferences, workshops, and hackathons manage registration, attendance, reminders, certificates, and feedback by hand — introducing delay, inconsistency, and zero post-event insight.

Siloed Registration

No unified participant record across tools or sessions

Unreliable Reminders

No automated targeting — attendees miss critical session updates

Slow Check-In

Manual check-in creates bottlenecks and inaccurate records

Certificate Bottlenecks

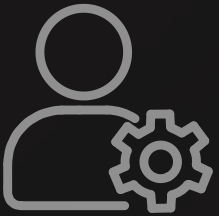
Delayed generation frustrates attendees and staff alike

Unanalyzed Feedback

Responses collected but never synthesized into actionable insight

Who This System Serves

Four distinct user groups, each with specific needs addressed by the automation platform.



Event Organizer / Admin

Needs a single dashboard to manage all event operations without manual overhead



Participant / Attendee

Needs seamless registration, timely reminders, and fast certificate delivery



Venue / Session Staff

Needs contactless, reliable check-in with real-time attendance visibility



Marketing / Community Team

Needs AI-generated feedback summaries and engagement analytics for future planning

What We Set Out to Automate

→ Registration & Record-Keeping

Auto-validate, de-duplicate, and store every participant record

→ Contactless QR Attendance

Scan-to-check-in with duplicate prevention and timestamp logging

→ Targeted Reminders

Session-specific emails sent once, only to the right attendees

→ Certificate Generation

Personalized PDFs created and delivered automatically post-event

→ AI-Driven Feedback Insight

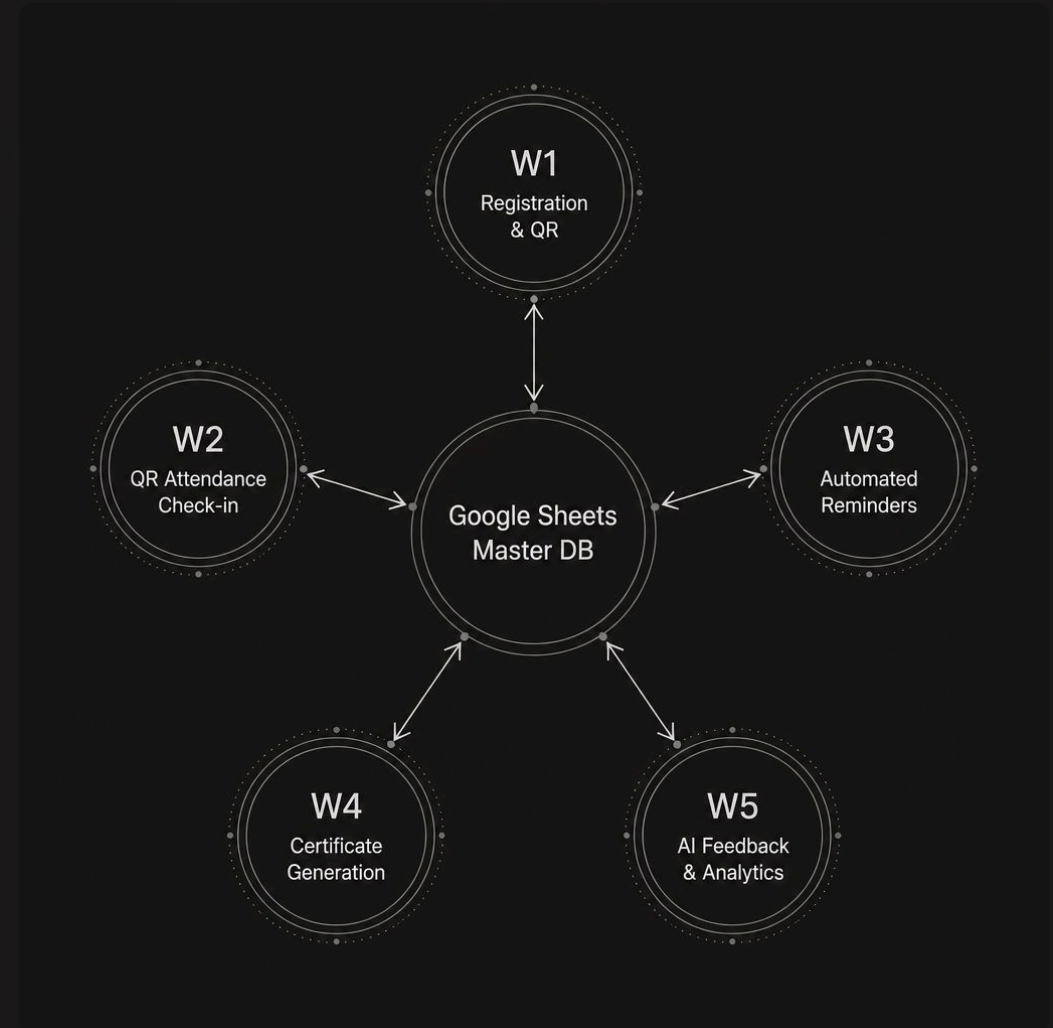
Sentiment and theme extraction without manual analysis

→ Full Audit Trail

Every action logged — scans, sends, errors, and retries tracked

Five Workflows, One Shared Database

A hub-and-spoke architecture with a Google Sheets **Master DB** as the single source of truth. All five n8n workflows read from and write to this shared layer — avoiding direct inter-workflow calls. The one physical link: the QR code generated in Workflow 1 becomes the trigger payload for Workflow 2.



Single Source of Truth

Google Sheets Master DB — all workflows share one data layer

Loose Coupling

Workflows operate independently, coordinated only through the shared DB

One Physical Link

QR code from W1 is the trigger payload consumed by W2's webhook

Registration → QR Code → Contactless Check-In

W1: Registration & QR

Trigger: New row added to Registration sheet

1. Validate input fields and de-duplicate records
2. Save participant record to Master DB
3. Generate unique QR code via QR API
4. Upload QR image to Google Drive
5. Send confirmation email with QR attached

W2: QR Attendance Check-In

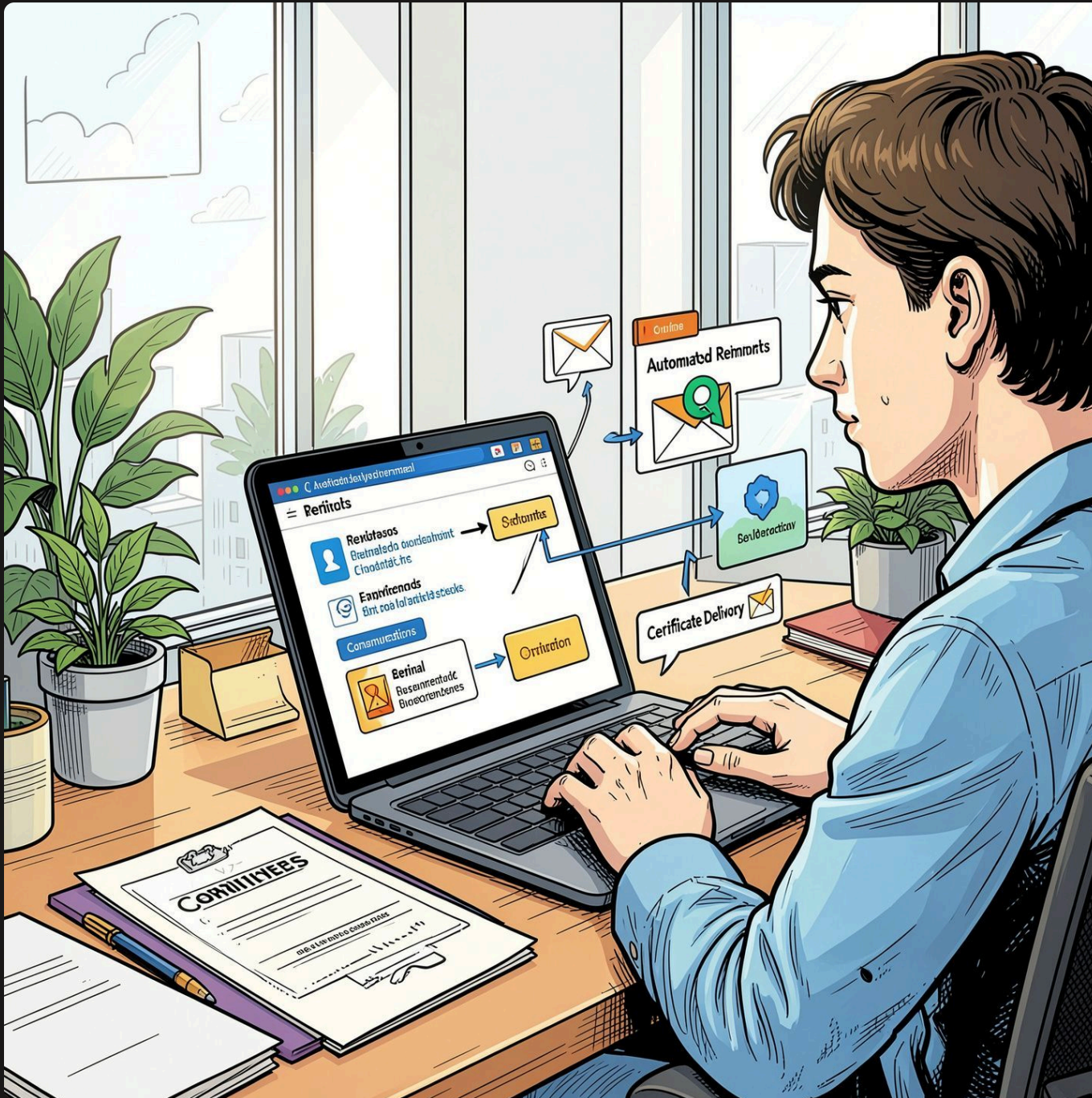
Trigger: Webhook fired when participant scans QR at venue

1. Validate participant record against Master DB
2. Check for duplicate check-in within same session
3. Mark attendance = TRUE, log timestamp
4. Log invalid or duplicate scans to audit trail

❏ Error handling includes retry logic for transient API failures and a dedicated audit log for all rejected scans.



Targeted Reminders & Automated Certificates



W3: Automated Reminders

Trigger: Hourly cron schedule — runs automatically without manual intervention

1. Read upcoming sessions from Master DB
2. Filter to sessions starting within 24 hours
3. Find participants not yet sent a reminder
4. Send personalized reminder via Gmail API
5. Mark reminder as sent in DB to prevent duplicates

W4: Certificate Generation

Trigger: Organizer-initiated webhook fired once event concludes

1. Filter attendees with attendance = TRUE and certified = FALSE
2. Generate personalized PDF via PDFMonkey API
3. Upload certificate to Google Drive participant folder
4. Email certificate with download link
5. Mark certified = TRUE in Master DB

☐ Built-in retry logic and status-polling ensures no participant is skipped due to transient API errors.

Feedback That Analyzes Itself



W5: AI Feedback & Analytics AI Capabilities

- Sentiment classification: Positive / Neutral / Negative
- Theme extraction across open-ended responses
- Aggregated summary report for stakeholder review

Real-Time Trigger

New feedback submitted → LLM extracts sentiment and theme instantly → results written to DB

Weekly Cron Trigger

LLM reads all feedback from past week → writes full analytics summary → summary emailed to organizers

Beyond Basic Automation



AI Decision Making

W5 — LLM-powered sentiment and theme extraction



Error Handling & Retry

W1, W2, W4 — transient API failure recovery built in



Logging & Audit Trail

W2 — all scans, sends, errors, and retries tracked



Scheduled / Cron Workflows

W3, W5 — hourly and weekly automated triggers



Webhook-Triggered Workflows

W2, W4 — real-time event-driven execution



Conditional Branching & Loops

All 5 workflows — dynamic routing and iteration

What This System Achieves

5

Workflows

Fully automated end-to-end

56

Total Nodes

Across all workflows

6

Integrations

Google Sheets, Drive, Gmail, PDFMonkey,
QR API, LLM API

Zero Manual Spreadsheet Work

After initial setup, all registration, attendance, and certificate operations run automatically

No Duplicate Sends or Certificates

DB-level flags prevent any participant from receiving duplicate reminders or certificates

Full Audit Trail

Every scan, send, error, and retry is logged and traceable

AI-Driven Insight Generated Automatically

Sentiment and theme analysis delivered to organizers without manual effort

Thank You!

